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ORIGINAL ARTICLE

Spinal tissue identification using a Forward-oriented endoscopic ultrasound technique

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Abstract
The limited imaging depth of optical endoscope restrains the identification of tissues under surface during the minimally invasive spine surgery (MISS), thus increasing the risk of critical tissue damage. This study is proposed to improve the accuracy and effectiveness of automatic spinal soft tissue identification using a forward-oriented ultrasound endoscopic system. Total 758 ex-vivo soft tissue samples were collected from ovine spines to create a dataset with four categories including spinal cord, nucleus pulposus, adipose tissue, and nerve root. Three conventional methods including Gray-level co-occurrence matrix (GLCM), Empirical Wavelet Transform (EWT), Variational Mode Decomposition (VMD) and two deep-learning based methods including Densely Connected Neural Network (DenseNet) model, one-dimensional Vision Transformer (ViT) model, were applied to identify the spinal tissues. The two deep learning methods outperformed the conventional methods with both accuracy over 95%. Especially the signal-based method (ViT) achieved an accuracy of 98.31% and a specificity of 99.2%, and the inference latency was only 0.0025 s. It illustrated the feasibility of applying the forward-oriented ultrasound endoscopic system for real-time intraoperative recognition of critical spinal tissues to enhance the precision and safety of minimally invasive spine surgery.

Keywords Ultrasound tissue characterization · Spinal tissue identification · Deep learning · Minimally invasive spine surgery

1 Introduction

The evolving lifestyle leads to an increasing of lumbar spine disorders. It was estimated that up to 80% of the population experiences spinal ailments at various stages in their lifetime [1]. Nowadays, traditional open spine surgeries have been gradually replaced by minimally invasive spine surgery (MISS), which can reduce surgical trauma and provide faster recovery. However, the primary challenge of current endoscopy lies in the limited intraoperative field of view (FOV) and operational space. Surgeons are required to perform sophisticated procedures such as discectomy and neural decompression through a surgical pathway less than 1 cm in width [2]. These procedures must be careful to avoid damaging critical tissues hidden behind the ligamentum flavum or adipose tissue, such as nerve roots and the spinal cord, otherwise any misjudgment can lead to significant injury.

Endoscopy Ultrasound (EUS) has served as a primary diagnostic tool for diseases such as choledocholithiasis, biliary microlithiasis and a variety of upper gastrointestinal tract tumors [3]. Moreover, EUS can characterize more details of tissues because of the rich physical information carried by ultrasound waves. Using various EUS techniques like elastography, targets required for biopsy can be precisely pinpointed [4]. However, there're limitations for the application of EUS in MISS. Firstly, the conventional EUS is usually equipped with a circular array, which is hard to operate in the limited operational space in MISS.

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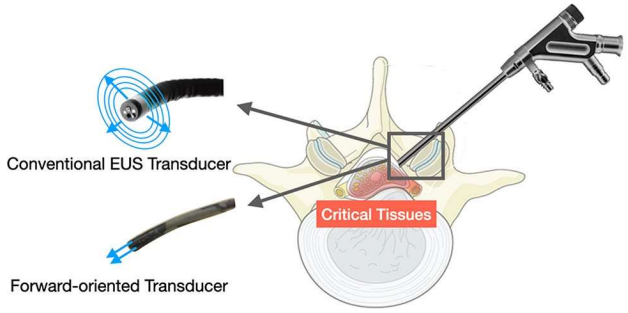


Fig. 1 The comparison of the conventional circular array transducer, and the proposed forward-oriented EUS transducer in MISS. The blue arrows represent the direction of US signals

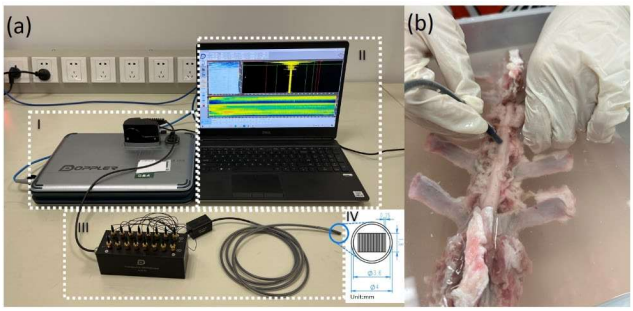


Fig. 2 The experimental setup: (a) The forward-oriented ultrasound endoscopic system: (I) the ultrasound detection board, (II) a laptop computer, (III) the miniature ultrasound array transducer and the adapter, (IV) the design sketch of the transducer, and the size was illustrated in mm; (b) Data acquisition from ovine spine samples in a water tank

Secondly, the circular array EUS can only provide the lateral information (Fig. 1), without the information along the normal direction, such as the tissue information of deeper layers along the surgical path, which is more important to the surgeons in MISS. Moreover, the lateral EUS remains a bad contact on an uneven area, which will reduce the image quality, increasing the risk of misdiagnoses. Several custom-designed forward ultrasound transducer probes have been developed for various scenarios. Chryssikos et al. [5] proposed a minimally invasive transducer with sized of 7 × 6 mm for imaging during anterior cervical spine surgery; Lv et al. [6] introduced a 26.2 MHz ultrasound transducer designed for high-precision interventional guidance. However, these transducers cannot meet the requirements of MISS because of their large size and small penetration depth caused by high frequency.

On the other hand, Ultrasonic Tissue Characterization (UTC) is a burgeoning field in the study of automatic tissue recognition, which utilizes the unique propagation characteristics of ultrasonic waves to recognize different tissues. A traditional paradigm in UTC involves extracting feature parameters from ultrasonic images or signals, followed by a machine learning classifier for tissue characterization. Huang et al. [7] developed a transient elastography (TE) system to compute the Young's modulus to assist in Liver Fibrosis Assessment, and the result showed a high correlation between conventional TE and the proposed system ($r = 0.975$). Kay Raum et al. [8] applied quantitative ultrasound (QUS) to measure the mesoscale stiffness tensor and elastic maps of the tissue for assessing bone quality. Quentin Grimal et al. [9] developed Resonant ultrasound spectroscopy (RUS) to measure the anisotropic elasticity of cortical bone, achieved a precision of 0.5% of measured Young and shear moduli. Georgas, et al. [10] applied shear wave ultrasound elastography (SWUE) to detect changes in cartilage mechanical properties for early detection of osteoarthritis. Ta et al. [11] used ultrasound backscatter signal to assess cancellous bone status of the neonates, which showed highly correlating with the birth days ($R = 0.633$). Bi et al. [12] used ultrasonic transmission technique with high-resolution peripheral quantitative computed tomography to assess bone loss. The ultrasonic transmission parameters exhibited moderate correlation of 0.61 with bone densities. Nowadays, deep learning has been applied to address UTC problem, such as breast cancer classification [13] and the grading of lung disease [14]. Yao et al. [15] proposed DeepThy-Net combining the information from ultrasound images and clinical factors to predict Cervical Lymph Node Metastasis in Papillary Thyroid Cancer and achieved an AUC between 0.870 and 0.905.

In this study, a forward-oriented ultrasound endoscopic system (Fig. 1a) was designed and utilized to maintain tight contact between transducer and spinal tissue surface, and to provide the tissue information of deep layer along the normal direction when collecting signals under limited FOV [16]. Two deep learning algorithms including Densely Connected Neural Network (DenseNet [17]) and one-dimensional Vision Transformer (ViT [18]) were applied for tissue identification tasks and then compared to conventional Gray-Level Co-occurrence Matrix (GLCM [19]), Empirical Wavelet Transform (EWT [20]) and Variational Mode Decomposition (VMD [21]) methods. Soft tissue samples in 4 categories as spinal cord (SC), nucleus pulposus (NP), adipose tissue (AT) and nerve root (NR) were acquired from ovine spines to validate the effectiveness and accuracy of the models.

2 Materials and methods

2.1 Forward-oriented scanning miniature ultrasound device

The experimental setup was depicted in Fig. 2. The miniature ultrasound (US) array transducer was custom-designed

and produced by local manufacturer (Doppler Electronic Technologies Co., Ltd., Guangzhou, China). The transducer array was composed by 16 piezoelectric elements which were made of 1-3-type composite piezoelectric material [22], and its diameter was 4 mm with the element pitch of 0.15 mm. The detailed parameters of transducer were the central frequency of 15 MHz, the bandwidth of 6 dB, the axial resolution of 51.3 μm, the lateral resolution of 0.52 mm, and the pulse duration of 0.3 μs. The acquisition system included an ultrasound detection board (Doppler Electronic Technologies Co., Ltd., Guangzhou, China) and an adapter which connected the transducer to the detection board, enabling the transmission and reception of US signals. All data were processed by a laptop computer (Intel(R) Core(TM) i7-10700 K CPU @ 3.80 GHz, 32 GB RAM).

In the experiment, the focus depth of the US transducer was set to 10 mm with a sampling frequency of 75 MHz and signal gain of 46 dB. The acquisition mode was set as synthetic aperture dynamic focusing with an aperture number of three elements, therefore each collection would record a 14-channel signal with 896 sampling points for each channel signal.

2.2 Data acquisition

Forty ovine spines purchased from local butcher shop were used for the ex-vivo experiments. The muscles and lamina were removed in order to mimic the surgical scenario and expose the anatomical structures such as SC, NR, NP and AT for signal acquisition. During the experiment, the spine samples were placed in a water tank, and the transducer was positioned on the surface of the tissues to collect signals (Fig. 2b).

A total of 758 radio frequency (RF) signals were collected for different soft tissues, including 223 from SC, 194 from NP, 109 from AT and 232 from NR. The number of training and testing datasets was set to 70% and 30%. To reduce the influence of the reflection from vertebral bone surfaces, the NP signals were manually truncated during preprocessing. In this study, ovine spines were used for the experiment, and the size of samples was smaller than human spine. Under the condition that the imaging depth was set to 10 mm during MISS, only reflection signals from the anterior part of the NP would be received, and the reflections from the vertebral bones would be excluded. However, the signals from ovine samples contained the reflection from vertebra bone surface due to small size, therefore should be truncated to match the real scenario during MISS.

As shown in Fig. 3, to convert the RF signals to images for further feature extraction, a Hilbert transformation was applied to the signals, followed by a logarithmic compression and a cubic interpolation to resize the images (224×224).

2.3 Algorithms

The UTC problem could be defined as: Given an ultrasonic sample (X, p), while X was the US image or signal, and p was the class label, the goal was to find a classifier, $f_{X \to p}$, which could predict the tissue label of the sample. Five methods were compared in this study including three conventional methods as GLCM, EWT and VMD, as well as two deep learning approaches as the image-based DenseNet and the signal-based ViT as shown in Fig. 4.

2.3.1 GLCM

GLCM [14] can extract the gray level statistical feature parameters for image texture analysis. In this study, four statistical properties were employed including contrast, angular second moment (ASM), entropy and inverse different moment (IDM).

Contrast reflects the change of local gray level and the sharpness of the image and can be calculated as

$$\text{Contrast} = \sum_{\text{i}} \sum_{\text{j}} (\text{i} - \text{j})^2 \text{p}(\text{i}, \text{j})$$

Angular Second Moment (ASM) Energy reflects the uniformity of image gray distribution and can be calculated as.

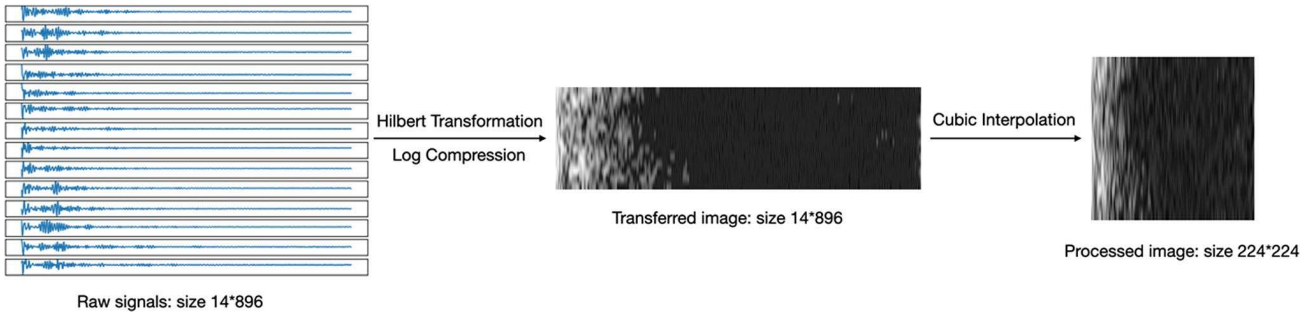


Fig. 3 The processing of converting RF signals to images

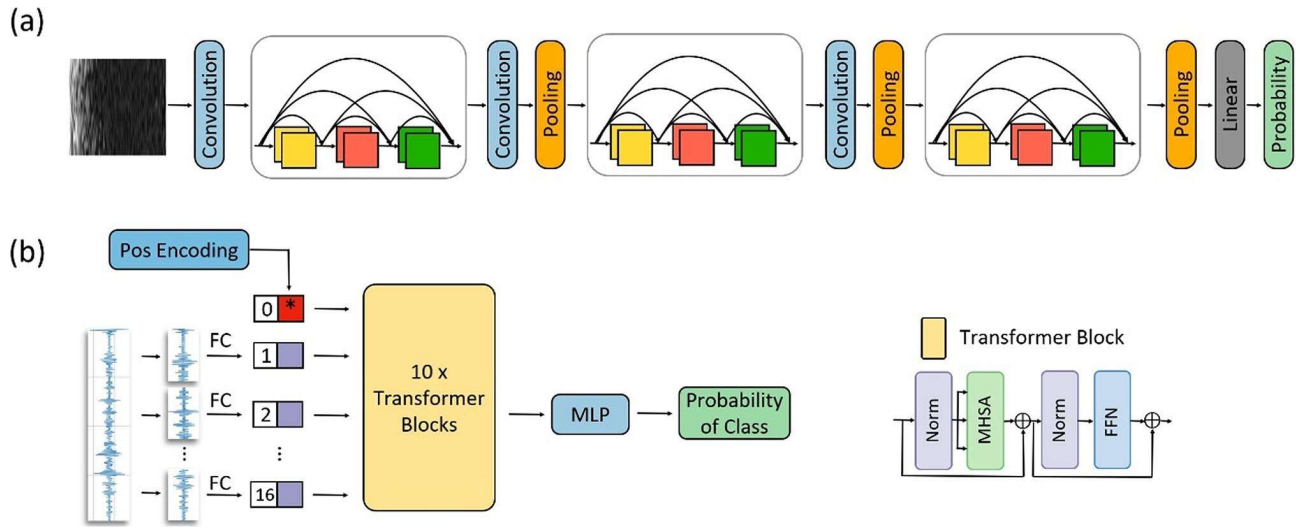


Fig. 4 The deep learning methods applied for tissue identification. (a) the image-based DenseNet model, and (b) the signal-based one-dimensional ViT

$$\mathrm{ASM} = \sum_{\mathrm{i}} \sum_{\mathrm{j}} \mathrm{p(i,j)}^2$$

Entropy is a characteristic parameter to measure the randomness of grayscale distribution and reflects the complexity of texture in the image. It can be calculated as

$$\mathrm{Entropy} = -\sum_{\mathrm{i}} \sum_{\mathrm{j}} \mathrm{p(i,j)} log\ \mathrm{p(i,j)}$$

Inverse Different Moment (IDM, Homogeneity) reflects the homogeneity of image texture, measures the local variation of image texture and can be calculated as

$$\mathrm{IDM} = \sum_{\mathrm{i}} \sum_{\mathrm{j}} \frac{1}{1+(\mathrm{i}-\mathrm{j})^2} \mathrm{p(i,j)}$$

where, (i, j) is the discrete coordinate of the image, and $0 \leq i \leq M-1, 0 \leq j \leq N-1$, M and N are the width and height; $p(i, j)$ represents the pixel intensity at(i, j). For each offset $(\Delta x, \Delta y)$, the properties are concatenated to a feature vector, which is sent to a support vector machine (SVM) for classification.

2.3.2 EWT

Empirical Wavelet Transform (EWT) [20] is a common conventional method to decompose 1D time series signal, and commonly employed to analyze ultrasonic radiofrequency (RF) echo signals. The EWT adaptively decomposes ultrasound signals into a series of empirical mode functions (EMFs), and the dominant EMFs are then identified for further analysis. The selected modes are processed to extract six statistical features—mean, variance, median, kurtosis, root mean square (RMS) and skewness—which are then used to form primary feature vectors for further classification.

2.3.3 VMD

Variational Mode Decomposition (VMD) [21] is a traditional signal processing method that is particularly useful for analyzing non-stationary and non-linear signals. It decomposes a signal into a predetermined number of band-limited Intrinsic Mode Functions (IMFs), with each IMF centered at a different frequency. The statistical features are extracted from each IMF to compose feature vectors to perform classification.

2.3.4 DenseNet

DenseNet [17] was selected as the classification model because of its high parameter efficiency and excellent generalization on small datasets. As shown in Fig. 4a, it introduced direct connections from any layer to all subsequent layers. Consequently, the lth layer received the feature maps of all preceding layers, x_0, x_1, ...,x_{l-1}, as input:

$$x_l = H_l\left([x_0, x_1, \ldots, x_{l-1}]\right)$$

where $[\mathrm{x_0}, \mathrm{x_1}, ..., \mathrm{x_{l-1}}]$ referred to the concatenation of the feature maps produced in layers 0, 1, …, l-1, and $\mathrm{H}(\bullet)$ was a composite function referring to three consecutive operations: batch normalization (BN), a rectified linear unit (ReLU) and a 3×3 convolution.

2.3.5 Vision transformer

Transformer is a widely used architecture in Nature Language Processing (NLP) due to its ability of processing sequential data. In this study, we adopted the idea from NLP and developed the Vision Transformer (ViT [18]) to process raw US signals. US signal is a type of sequential data, whose reflection patterns can be analyzed by a transformer. Although US signals and sentences in NLP are both one-dimensional sequences, they are quite different on the tokenization process. The tokens of a sentence are based on a pre-defined vocabulary, which makes them clear and meaningful; while the sampling points of the signal are noisy and discrete. Therefore, it's unsuitable to tokenize US signals by each sampling point; instead, we tokenized the US signal by groups of sampling points.

Specifically, as shown in Fig. 4b, the US signal was split into several patches, and a fully connected layer was applied to tokenize the signal into a token sequence. After adding a class token and a positional encoding, the features were fed into transformer blocks to learn attention scores. The transformer blocks were comprised of a multi-head self-attention (MHSA) module and a Feed-Forward Network (FNN) consistent to the transformer encoders in NLP [23]. Finally, a multi-layer perceptron (MLP) was connected to the end of the model to output the predictions.

2.4 Implementation details & evaluation metrics

For GLCM, three groups of offsets, $(0, 0)$, $(0, 1)$, $(1, 0)$, were applied to calculate textural features of three directions. An SVM with Gaussian kernel was employed to classify the feature vectors.

For EWT and VMD, each signal was decomposed into four parts to extract statistical features. A K-Nearest Neighbors (KNN) classifier was utilized to achieve classification.

For DenseNet and ViT, all code implementations were based on PyTorch library. The objective function was the Cross Entropy, and the optimization algorithm was AdamW. The learning rate was dynamically adjusted by CosineAnnealingLR scheduler of PyTorch.

The model performance was evaluated by accuracy (Acc), precision (Prec), sensitivity (Sens), specificity (Spec) and F1 scores (F1)[24]. Inference latency represents the time required for predicting one signal sample. All methods were conducted on the same Linux server, equipped with a NVIDIA RTX3090 GPU with 24 GB RAM.

3 Results

The examples of hematoxylin-eosin staining (H&E) images, ultrasound signals and images for four tissue types were illustrated in Fig. 5. The SC (Fig. 5a) was wrapped in a dura mater spinalis which can be considered as dense interfaces. The ultrasound sequentially penetrated the first and second layers of the dura mater spinalis in the scanning direction, therefore the US signal and image exhibited strong reflection at depths of 1 mm and 8 mm. The NP (Fig. 5b) and AT (Fig. 5c) were both soft tissues, presenting relatively homogeneous structure in the H&E images. Comparing to AT, NP had denser textural features, which exhibited more texture contents in the signal and image as shown in the boxes of Fig. 5b. The NR (Fig. 5d) contained numerous nerve fibers that emanated from the spinal cord, thus the US signals and images appeared periodical structures which indicated the network-like structures of nerve root.

The statistical results of tissue identification were demonstrated in Table 1. Comparing to conventional algorithms, the learning-based DenseNet and ViT showed improved performance with both accuracy higher than 95%. However, DenseNet still showed more false prediction of SC and NR which were misdiagnosed as AT. The signal-based ViT could recognize spinal soft tissues more effectively with highest accuracy of 98.31%. Table 1 listed five evaluation metrics and inference latency for the three methods. All metrics for ViT exceeded 96%, indicating that the signal-based method showed high recognition capabilities for spinal soft tissues. Furthermore, ViT achieved an inference latency of 0.0025 s on a single GPU, which indicated that the signal-based method would reduce 95% computing time than DenseNet.

4 Discussion

Two learning-based methods, DenseNet and ViT, had outperformed three traditional methods with higher accuracy. Deep learning is necessary in this case because it could capture more information from ultrasound images or signals which would not be identificated by visual observation or conventional signal processing. Figure 6 illustrated two ultrasound signals from NR and SC. Visually, the two signals presented similar features, and traditional methods misclassified the signal from NR (Fig. 6a) as SC, however the deep learning method correctly identified it as NR. This example depicted the ability of deep learning to detect subtle features that were not apparent to traditional methods, leading to more precise detection.

Three signal-based methods showed higher accuracies than the two image-based methods. For traditional methods,

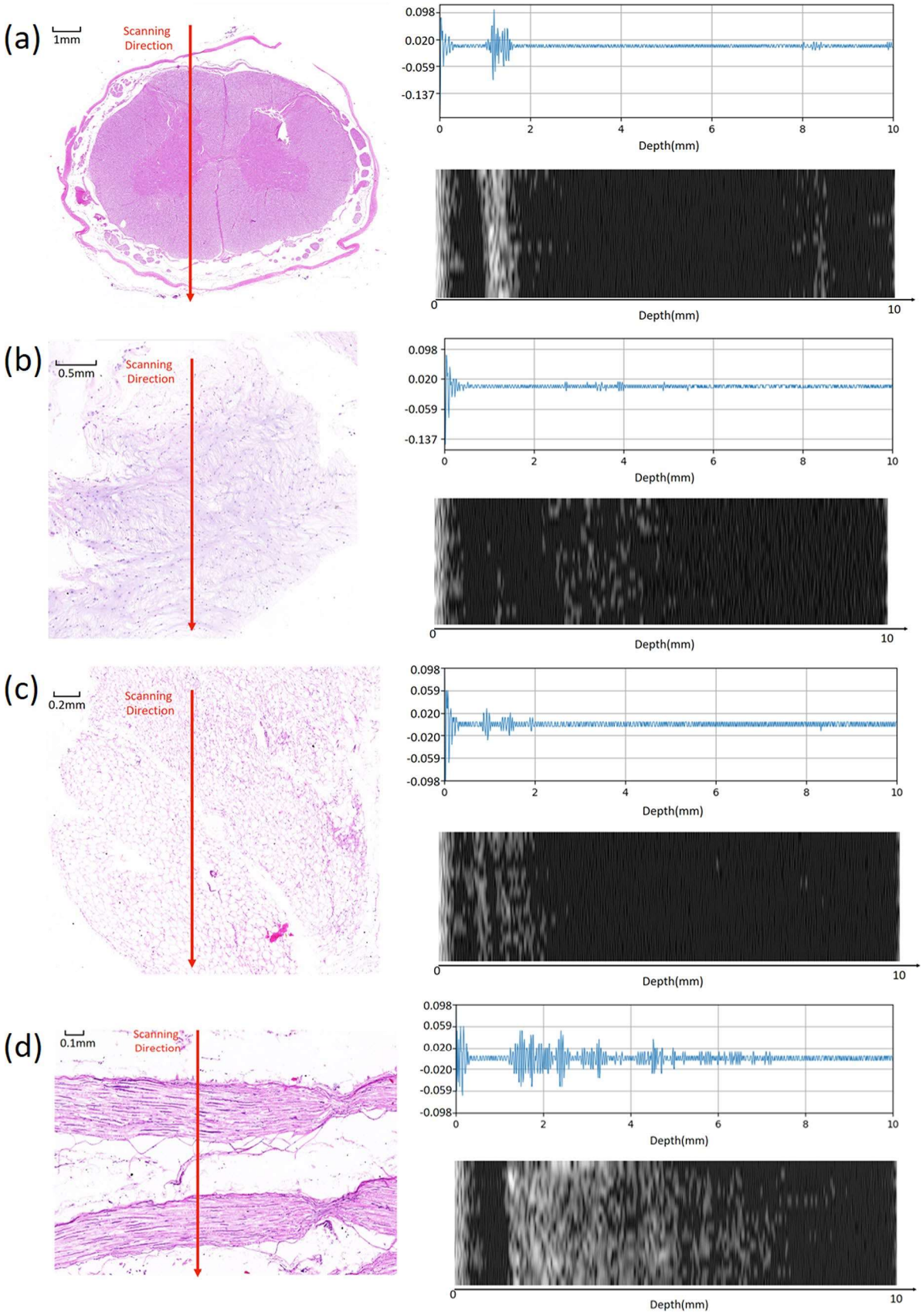


Fig. 5 The H&E images, US signals and US images collected from: (a) Spinal Cord (SC), (b) Nucleus Pulposus (NP), (c) Adipose Tissue (AT), (d) Nerve Root (NR)

Table 1 Statistical results of spine soft tissue identification

Method	Acc (%)	Prec (%)	Sens (%)	Spec (%)	F1 (%)	Latency (s)
GLCM	79.92	61.68	62.81	86.89	59.71	0.1443
EWT	83.26	83.32	76.64	94.14	79.84	0.0255
VMD	90.22	89.72	87.20	96.60	88.44	0.1024
DenseNet	95.14	92.73	89.01	89.91	95.68	0.0598
ViT	98.31	96.46	96.63	99.20	96.38	0.0025

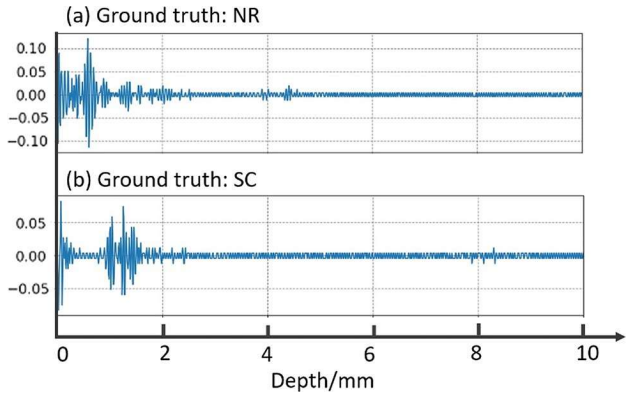


Fig. 6 The ultrasound signals with similar visual features from (a) NR and (b) SC

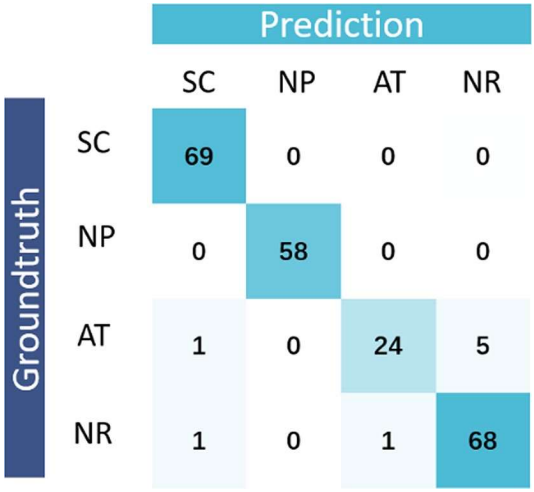


Fig. 7 The Confusion matrix of ViT

two conventional signal-based methods, EWT and VMD, were utilized for the identification of four soft tissues in the spine. The accuracy of EWT reached 83.26%, while VMD achieved the highest classification accuracy among the three traditional methods at 90.22%. Both EWT and VMD exhibited better performance in classifying than the conventional image-based GLCM. For learning-base methods, The images-based DenseNet showed slightly lower sensitivity and specificity than the signal-based ViT, since some useful classification information was lost during the process of converting signals to image. Ultrasound signal is a kind of time series, therefore the UTC problem can be treated as a Time Series Classification (TSC) problem. It has been approved that the signal-based classification deep networks such as InceptionTime[25], FEDFormer[26], and TimesNet [27] showed high excellency to address TSC problems. In this study, ViT could extract more detailed features from US signals in time and frequency, and thus indicated the best performance over five methods.

Our previous study [16] employed GLCM followed by a K-nearest-neighbor classifier to identify 6 classes of spinal tissues and achieved an overall accuracy of 73.3%. However, the average identification accuracy of soft tissues was only 60%. Specifically, the accuracy of SC and NP was 60%, and AT was only 20%. In this study, bone and annulus fibrosus were removed from the identification task, because these hard tissues could be easily distinguished from other 4 soft tissues simply by observation during MISS. Selecting four types of soft tissues as classification target can make the training model focus on subtle features and improve the performance, and additionally more in line with the practical scenarios. The results illustrated that the accuracy for soft tissues was improved to 79.92% even only applying traditional GLCM.

For ViT, there were still 5 false predictions of AT mistakenly predicted as NR, as shown in Fig. 7. One example case was depicted in Fig. 8a. The AT was expected to show weak reflections (Fig. 8b), but in this case, the signal showed a periodic pattern, which made it look like NR tissue (Fig. 8c). It may result from the partial or loose contact between transducer and tissue surface, which can distort the reflection of ultrasound signals and increase the chance of misdiagnosis. The frozen and unfrozen storage process of the ovine spine could also change the micro-structure of the AT, leading to the wrong prediction. Although the accuracy of AT was lower than other tissues, doctors could easily identificate AT by visually ensuring in surgery because the surface of AT was ivory or yellowish, which was different from other tissues such as the white and smooth thecal sac. Our further work will implement multi-modality method such as combining optical imaging and US imaging to improve the accuracy of spine soft tissue identification.

Comparing the latency of five methods in Table 1, the ViT exhibited the shortest latency of 0.0025 s. This was attributed to ViT's self-attention mechanism, which modeled signals more effectively, thereby enhancing processing

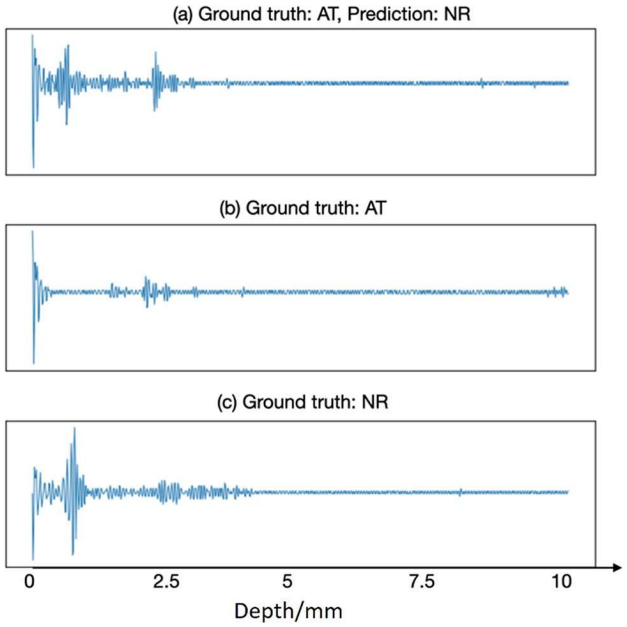


Fig. 8 A wrongly predicted case using ViT, (a) The RF signal of AT which was wrongly predicted as NR, (b) an example RF signal of AT, (c) an example RF signal of NR

speed and accuracy. This is particularly significant for real-time tissue identification during surgery.

In this study, the experiments were confined to ex-vivo tests on ovine spines and not fully accorded with the conditions of clinical scenarios. The in-vivo experiments will be implemented to validate the applicability of ViT in the future, and the experiments in a limited FOV will be performed. Meanwhile, our future work will modify the design of miniature transducer to keep good contact more easily during acquisition so as improving the identification performance.

5 Conclusion

It was demonstrated that the proposed signal-based ViT was an effective method to distinguish the patterns of ultrasound signals collected from the forward-oriented endoscopic ultrasound system, and could precisely and efficiently identify different spinal soft tissues under the limited operational space and intraoperative FOV. The inference latency of 0.0025 s could meet the requirements for real-time tissue recognition during minimally invasive spine surgery. The proposed system with application of the UTC algorithm has shown the feasibility to assist surgeons for real-time tissue identification during endoscopic spine surgery, furthermore enhancing surgical precision and safety.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Jiaqi Yao, Chang Jiang, Yiwei Xiang, Zhiyang Zhang, and Zixian Chen. The algorithms were designed and implemented by Jiaqi Yao, Yiwei Xiang, Rui Zheng, and Fei Gao. The first draft of the manuscript was written by Jiaqi Yao, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Conflict of interest No conflict of interest.

Ethical approval Not involved.

Consent to participate Not involved.

Consent for publication Not involved.

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